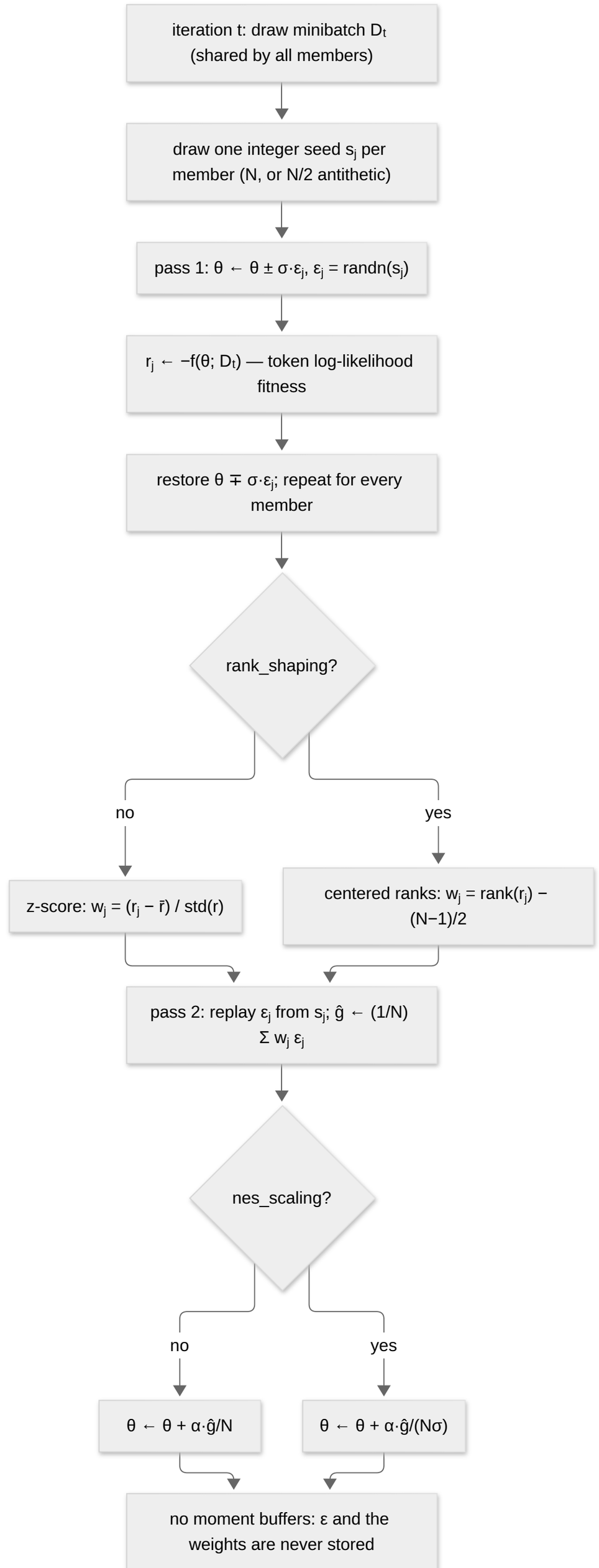

iteration t: draw minibatch D_t (shared by all members)
draw one integer seed s_j per member (N, or N/2 antithetic)
pass 1: $\theta \leftarrow \theta \pm \sigma \cdot \varepsilon_j$, ε_j = randn(s_j)
$r_j \leftarrow -f(\theta; D_t)$ — token log-likelihood fitness
restore $\theta \mp \sigma \cdot \varepsilon_j$; repeat for every member
rank_shaping?
no
yes
z-score: $w_j = (r_j - \bar{r}) / \mathrm{std}(r)$
centered ranks: $w_j = \mathrm{rank}(r_j) - (N-1)/2$
pass 2: replay ε_j from s_j; $\hat{g} \leftarrow (1/N) \Sigma\, w_j\, \varepsilon_j$
nes_scaling?
no
yes
$\theta \leftarrow \theta + \alpha \cdot \hat{g}/N$
$\theta \leftarrow \theta + \alpha \cdot \hat{g}/(N\sigma)$
no moment buffers: ε and the weights are never stored